

Diagnosing Reward Misspecification and Observation Insufficiency from Finite Prefixes of Reinforcement-Learning Training

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Abstract

We ask when a finite prefix of ordinary reinforcement-learning (RL) training identifies whether poor intended performance is caused by optimizing a misspecified proxy reward or by lacking task-relevant observations. We define the two failures as separate value gaps, distinguish intrinsic missing information from limited agent memory, and formalize finite-prefix identifiability as a property of distributions over passive logs. A two-action construction proves that passive logs can be exactly observationally equivalent under the two failure mechanisms. We then define privileged-state and reward-audit probes and pose selection of a diagnostic intervention as a minimum-cost statistical decision problem. The specification yields explicit ground-truth labels and testable estimands for the accompanying experiments.

1 Research question and scope

Research question. When are reward misspecification and task-relevant observation insufficiency statistically identifiable from a finite prefix of RL training, and, when passive logs are insufficient, what minimum-cost diagnostic intervention makes them identifiable?

Reward misspecification can allow greater proxy return while reducing intended return [1]. Partial observability creates a different difficulty: an agent must act without full knowledge of the controlled latent state, and tractability requires structural conditions on how observations reveal that state [2]. Early interactions are worth studying because they can exert a lasting influence on deep-RL learning [3]. Our object is not merely prediction of eventual failure, but identification of its causal class from early data.

2 Formal setting

Let a finite-horizon POMDP with two objectives be

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, Z, \mu_0, H, \gamma, \tilde{r}, r^*). \quad (1)$$

Here $S_0 \sim \mu_0$, $O_t \sim Z(\cdot \mid S_t)$, $S_{t+1} \sim P(\cdot \mid S_t, A_t)$, $\tilde{r} : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is the reward supplied during ordinary training, and $r^* : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is the designer’s intended evaluation objective. The analysis also permits stochastic rewards by interpreting these functions as conditional means. For objective r and policy π , define

$$V_r(\pi) = \mathbb{E}_{\mathcal{M}, \pi} \left[\sum_{t=0}^{H-1} \gamma^t r(S_t, A_t) \right]. \quad (2)$$

Three policy classes isolate information from computation:

Π_{full} : policies that observe the current latent state S_t ,

Π_{hist} : policies with perfect recall of the observable history,

Π_m : policies restricted to a specified memory budget m .

The observable history before action t is $H_t^o = (O_0, A_0, \tilde{R}_0, \dots, O_t)$; thus Π_{hist} may use all information actually available to an ideal agent, whereas $\Pi_m \subseteq \Pi_{\text{hist}}$ represents the architecture being trained (for example, a reactive policy or a fixed recurrent state).

For a learning algorithm A , seed U , and T training transitions, the *passive early-training prefix* is

$$\mathcal{L}_T = (O_t, A_t, \tilde{R}_t, p_t, V_t, \delta_t, d_t)_{t < T}, U, \quad (3)$$

where p_t is the logged action distribution, V_t a value estimate, δ_t a TD error, and d_t an episode boundary indicator. Latent states S_t and intended rewards R_t^* are deliberately excluded. Quantities such as entropy or action diversity are measurable functions of $\mathcal{L}_T$ and therefore add no information in the statistical sense, although they may make classification easier for a finite-capacity diagnostic model.

3 Failure definitions

For a policy class Π , let $J_r(\Pi) = \sup_{\pi \in \Pi} V_r(\pi)$ and $\Gamma_r(\Pi) = \arg \max_{\pi \in \Pi} V_r(\pi)$.

Definition 1 (Reward misspecification gap). *Holding the information interface fixed at Π_{hist} , define*

$$\Delta_R = J_{r^*}(\Pi_{\text{hist}}) - \sup_{\pi \in \Gamma_r(\Pi_{\text{hist}})} V_{r^*}(\pi). \quad (4)$$

Thus $\Delta_R > 0$ means that even the intended-best proxy-optimal policy is suboptimal for the intended objective. For an actual learned policy $\hat{\pi}_T$, the empirical, algorithm-dependent regret is $\hat{\Delta}_{R,T} = J_{r^*}(\Pi_{\text{hist}}) - V_{r^*}(\hat{\pi}_T)$ after controlling for the observation gap below.

This definition is behavioral: $\tilde{r} \neq r^*$ alone is insufficient, because two rewards may induce the same optimal policies. It also avoids labeling an arbitrary unfavorable tie as structurally unavoidable.

Definition 2 (Intrinsic observation gap).

$$\Delta_O = J_{r^*}(\Pi_{\text{full}}) - J_{r^*}(\Pi_{\text{hist}}). \quad (5)$$

If $\Delta_O > 0$, even an ideal policy with unlimited recall of the available history cannot match a state-informed policy; task-relevant information is intrinsically absent from the observation process.

Definition 3 (Memory/architecture gap).

$$\Delta_M^{(m)} = J_{r^*}(\Pi_{\text{hist}}) - J_{r^*}(\Pi_m). \quad (6)$$

A positive $\Delta_M^{(m)}$ is not, by itself, observation insufficiency: the required information exists in the observable history but the chosen agent cannot retain or use it. This gap can be estimated by comparing a history-aware oracle (or sufficiently expressive recurrent policy) with the trained architecture.

Given margins $\varepsilon_R, \varepsilon_O > 0$, assign the population label

$$Y(\mathcal{M}) = (\mathbf{1}\{\Delta_R > \varepsilon_R\}, \mathbf{1}\{\Delta_O > \varepsilon_O\}) \in \{0, 1\}^2, \quad (7)$$

corresponding to neither, reward-only, observation-only, and both. Margins prevent negligible numerical differences from defining a failure.

4 Finite-prefix identifiability

Let Θ be a prespecified family of environment–objective pairs and let P_θ^T denote the distribution of $\mathcal{L}_T$ induced by $\theta \in \Theta$, the fixed learner, and the seed distribution.

Definition 4 (Passive identifiability at prefix length T). *The label is passively identifiable on Θ if*

$$P_\theta^T = P_{\theta'}^T \implies Y(\theta) = Y(\theta') \quad \forall \theta, \theta' \in \Theta. \quad (8)$$

Equivalently, every observational-equivalence class induced by the passive prefix is label-homogeneous.

For two simple hypotheses with equal prior probability, the optimal diagnostic error after n independent runs is

$$e_n^* = \frac{1 - \text{TV}((P_0^T)^{\otimes n}, (P_1^T)^{\otimes n})}{2}. \quad (9)$$

Hence identical prefix distributions force chance-level binary diagnosis for every sample size. Merely adding deterministic summaries of the same logs cannot break this equivalence.

4.1 An explicit passive impossibility result

Fix $q \in (1/2, 1)$ and consider independent one-step episodes with latent $S \in \{0, 1\}$, $\mathbb{P}(S = 0) = q$, actions $A \in \{0, 1\}$, and the constant observation $O = o$. Define two systems:

$$\begin{aligned} \mathcal{M}_O : \quad & \tilde{R} = R^* = \mathbf{1}\{A = S\}; \\ \mathcal{M}_R : \quad & R^* = \mathbf{1}\{A = 1\}, \\ & \tilde{R} \mid A = a \sim \begin{cases} \text{Bernoulli}(q), & a = 0, \\ \text{Bernoulli}(1 - q), & a = 1, \end{cases} \end{aligned}$$

where $\tilde{R} \perp S \mid A$ in $\mathcal{M}_R$.

Proposition 1 (Passive non-identifiability). *For any adaptive learner whose action at each episode is a measurable function of its seed and previous passive logs,*

$$P_{\mathcal{M}_O}^T = P_{\mathcal{M}_R}^T \quad \text{for every finite } T. \quad (10)$$

Nevertheless, $\mathcal{M}_O$ is observation-limited but not reward-misspecified, while $\mathcal{M}_R$ is reward-misspecified but not observation-limited.

Proof sketch. In $\mathcal{M}_O$, marginalizing the unobserved state gives $\tilde{R} \mid A = 0 \sim \text{Bernoulli}(q)$ and $\tilde{R} \mid A = 1 \sim \text{Bernoulli}(1 - q)$, exactly as in $\mathcal{M}_R$. Because O is constant, induction over episodes shows equality of the full passive-log distributions under any adaptive action rule. Logged policy probabilities, value estimates, and TD errors are produced by the same learner from the same seed and history, so their distributions are also equal.

In $\mathcal{M}_O$, proxy and intended rewards coincide, so $\Delta_R = 0$. A full-state policy chooses $A = S$ and obtains value 1, while the best history policy always chooses $A = 0$ and obtains q ; therefore $\Delta_O = 1 - q > 0$. In $\mathcal{M}_R$, the unique proxy-optimal action is 0, whereas the intended-optimal action is 1, so $\Delta_R = 1$. State access cannot improve the intended objective, which is independent of S , and thus $\Delta_O = 0$. $\square$

The construction also rules out explanations based only on reward scale: the entire conditional distribution of the logged proxy reward is matched, not just its mean.

5 Diagnostic interventions

An intervention d modifies only the diagnostic data-collection protocol, not the population label. We consider:

- (i) **Passive** (d_O): collect additional ordinary episodes.
- (ii) **Privileged state** (d_S): reveal S_t to a diagnostic probe for selected decisions, while leaving the training reward unchanged.
- (iii) **Reward audit** (d_R): query R_t^* on selected realized or counterfactual state–action pairs, or run short probes with corrected reward.
- (iv) **Combined** (d_{SR}): permit both forms of privileged access.

The interventions target different causal contrasts. A state probe estimates the gain $J_{r^*}(\Pi_{\text{full}}) - J_{r^*}(\Pi_{\text{hist}})$; a reward audit estimates how the intended ranking of actions differs from the proxy ranking. In the toy pair, privileged state predicts proxy reward in $\mathcal{M}_O$ but is irrelevant to proxy reward in $\mathcal{M}_R$, whereas an audit finds $R^* = \tilde{R}$ in $\mathcal{M}_O$ but not in $\mathcal{M}_R$. Either targeted probe therefore breaks the passive equivalence statistically; using both is needed for unrestricted four-class diagnosis when the two failures may coexist.

Let N_d be probe episodes, $Q_S(d)$ privileged-state queries, and $Q_R(d)$ intended-reward queries. For application-dependent nonnegative prices, define

$$C(d) = c_E N_d + c_S Q_S(d) + c_R Q_R(d). \quad (11)$$

Writing $P_\theta^{T,d}$ for all passive and interventional data, the minimum-cost δ -reliable intervention is

$$C_\delta^* = \inf_{d,g} \left\{ C(d) : \sup_{\theta \in \Theta} \mathbb{P}_\theta[g(\mathcal{L}_T, D_d) \neq Y(\theta)] \leq \delta \right\}. \quad (12)$$

Here g ranges over diagnostic rules and D_d is the probe data. A necessary pairwise condition is

$$\inf_{Y(\theta) \neq Y(\theta')} \text{TV}(P_\theta^{T,d}, P_{\theta'}^{T,d}) \geq 1 - 2\delta. \quad (13)$$

Thus the “minimum” intervention is not universally one query; it depends on the candidate family, noise level, confidence target, and relative cost of state and reward access.

6 Assumptions for positive identification

The impossibility result shows that identification requires restrictions or interventions. The initial experiments should state and test the following:

- A1. Known protocol.** The learner, logging map, prefix length, and seed distribution are fixed or observed; train and diagnostic trajectories do not silently change across conditions.
- A2. Exploration/overlap.** Every action needed to distinguish the hypotheses is selected with probability at least $\eta > 0$ in relevant histories, or is deliberately selected by a probe.
- A3. Margin and stationarity.** Failure gaps exceed their declared thresholds, and transition, observation, and reward mechanisms remain stable during a run.
- A4. Faithful probes.** Privileged state reveals task-relevant state without changing dynamics; the reward audit is an unbiased measurement of the declared intended objective.
- A5. Structural separation.** Within the chosen environment family, every cross-label pair is separated by the selected passive or interventional data distribution. Without this assumption, no classifier can guarantee causal identification.

Under A1–A5 and independent runs, standard consistent hypothesis tests recover the label as the number of runs grows. Finite-sample claims should be reported with uncertainty rather than inferred from a small set of point estimates; this is especially important in few-run RL evaluation [4].

7 Experimental design and implementation

The implementation stores S_t , R_t^* , and $\tilde{R}_t$ separately, exposing only the latter in passive logs. Ground-truth labels live in a separate manifest keyed by opaque run identifiers. The following oracle comparisons validate the ground-truth condition before training a diagnostic:

Quantity	Population contrast	Interpretation
Δ_R	intended-best vs. proxy-optimal	incentive failure
Δ_O	full-state vs. full-history	missing information
$\Delta_M^{(m)}$	full-history vs. memory- m	agent limitation

Controlled environments. The factorial benchmark has two latent states and two actions, horizon $H = 5$, and $\mathbb{P}(S = 0) = 0.75$. The intended reward is $R^* = \mathbf{1}\{A = S\}$. Observation corruption maps both states to a sentinel; reward corruption changes the proxy target to $1 - S$. The four independent toggle settings realize neither, reward-only, observation-only, and both failures. A second benchmark implements the exact two-system construction in the proposition. It is essential: the factorial benchmark is a controlled positive task, whereas only the matched pair tests passive impossibility.

Learners, logging, and probes. We ran tabular Q-learning and clipped tabular PPO for 1,000 transitions using five paired seeds in each condition. Across four factorial conditions and two matched controls this gives 60 runs and 60,000 passive transitions. Passive rows contain observations, actions, proxy rewards, action probabilities, entropy, value estimates, TD errors, boundaries, steps, and seeds; neither latent state, intended reward, nor class label is present. State, reward-audit, and combined probes use balanced actions and 4, 8, 16, 32, or 64 probe episodes. Prices are $c_E = 1$ and $c_S = c_R = 4$.

Diagnostics and splits. We extract the first 5%, 10%, and 20% of every run. Baselines use random guesses, reward-curve summaries, or standard training statistics (entropy, action diversity, observation coverage, TD error, episode length, and a

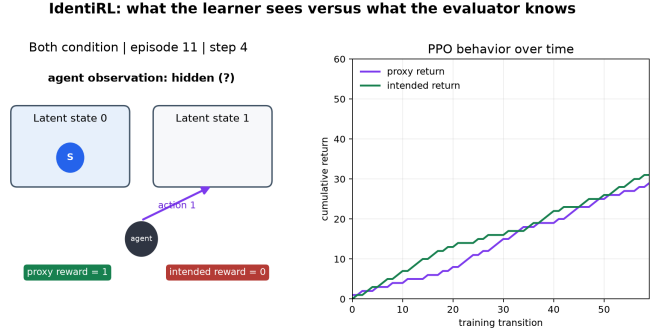


Figure 1: A frame from the reproducible 2D playback. The learner sees the sentinel observation but the evaluator separately records latent state and intended reward. Here the proxy rewards action 1 even though intended reward is zero, directly visualizing the “both” condition.

held-out history-dependence score). Logistic diagnostic models use leave-one-paired-seed-out splits, so no transition from a held-out trajectory enters training. We report balanced accuracy, macro-F1, class recall, confusion matrices, and paired-seed bootstrap intervals.

8 Empirical results

Ground-truth gaps. With $q = 0.75$ and $H = 5$, the factorial reward gaps are 0, 5, 0, and 2.5 for neither, reward-only, observation-only, and both; observation gaps are 0, 0, 1.25, and 1.25. The matched observation system has gaps (0, 1.25) and the matched reward system (5, 0). The memory gap is zero throughout: useful information is either current or intrinsically absent. Every reward has binary support $[0, 1]$, ruling out reward scale as a shortcut.

Passive impossibility check. The empirical proxy-reward rates conditioned on action agree closely across the two matched mechanisms. For PPO, the absolute differences are 0.0045 for action 0 and 0.0023 for action 1; for Q-learning they are 0.0092 and 0.0113. This is sampling fluctuation around exact equality in population. Accordingly, passive standard-statistics accuracy on the matched control is 0.50 for PPO and 0.40 for Q-learning. Finite-sample scores above or below chance do not contradict non-identifiability.

Interventions and cost. Combined probes reach perfect balanced accuracy in all four learner–benchmark combinations at 64 episodes. Using the conservative rule that the 95% bootstrap lower bound must be at least 0.90, the cheapest tested intervention at the 20% prefix costs 20 for both factorial learners (four state or reward probe episodes), 40 for PPO on the matched control (eight reward audits), and 72 for Q-learning on the matched control (eight combined probes). These minima are specific to this candidate family, cost vector, and five-seed sample.

Table 1: Balanced diagnostic accuracy at the 20% prefix. Probe columns use 64 episodes. Chance is 0.25 for factorial diagnosis and 0.50 for the matched binary control.

Learner	Benchmark	Random	Passive stats	State	Reward audit	Combined
Q-learning	Factorial	0.35	0.45	1.00	0.80	1.00
Q-learning	Matched	0.30	0.40	0.90	0.80	1.00
PPO	Factorial	0.35	0.70	1.00	0.95	1.00
PPO	Matched	0.30	0.50	1.00	1.00	1.00

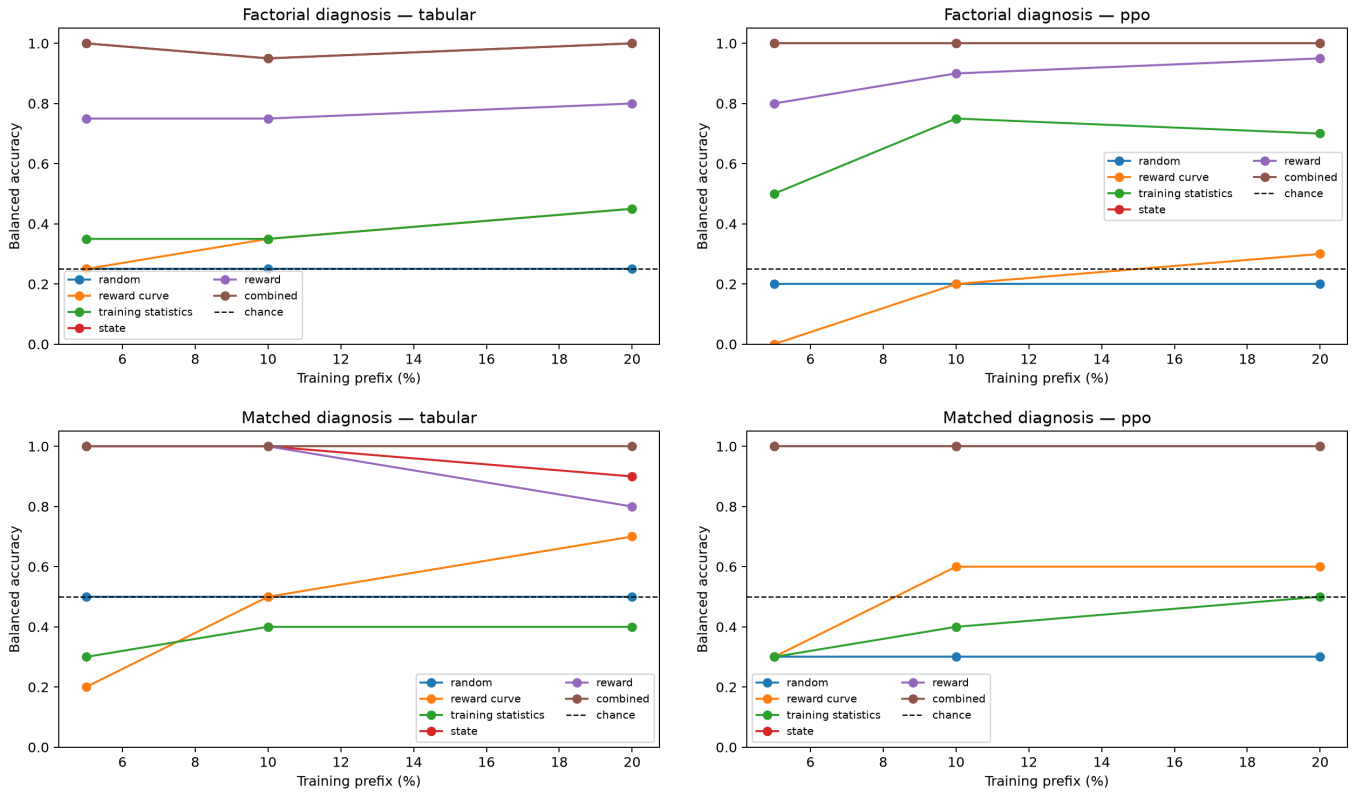


Figure 2: Balanced accuracy against passive-prefix length. The matched control keeps standard passive diagnosis near chance, while targeted probes separate the mechanisms. Probe curves shown here use 64 episodes.

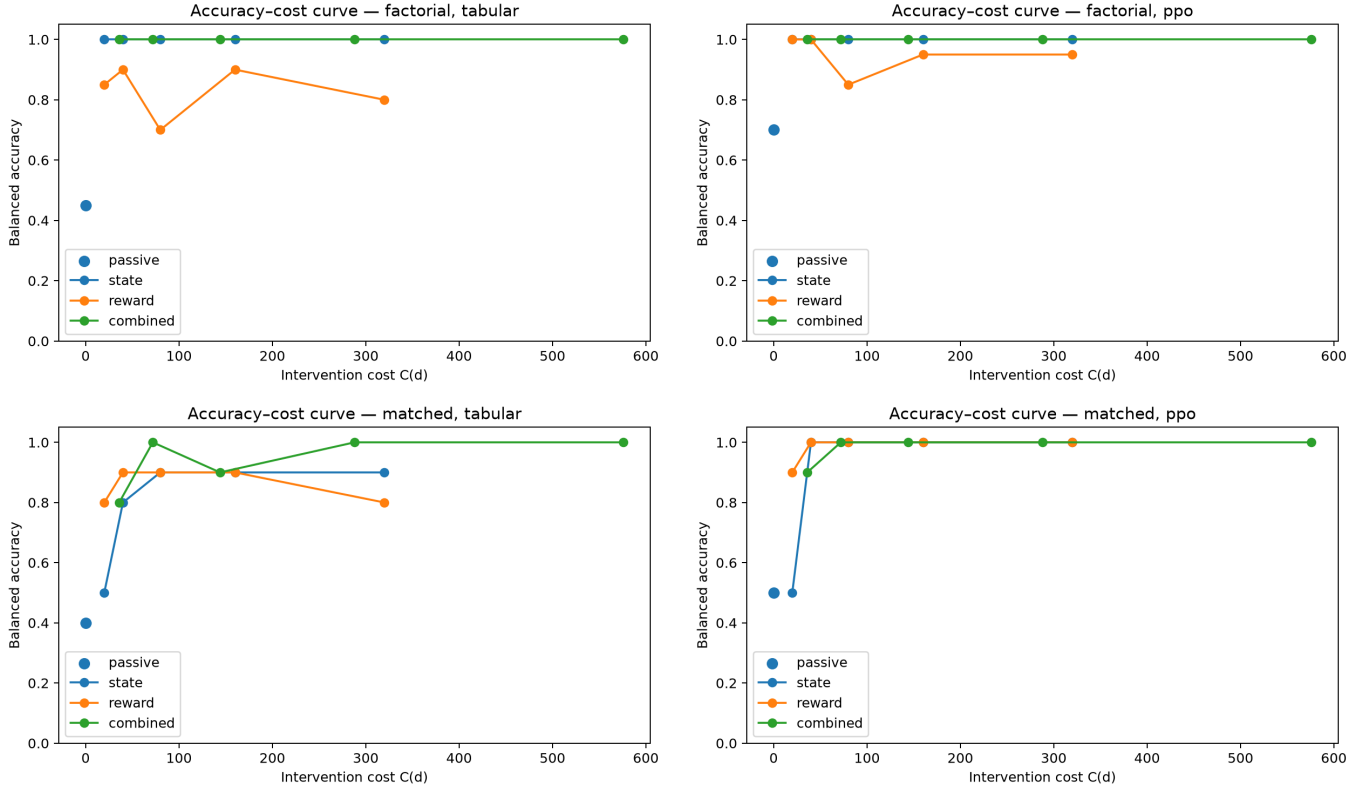


Figure 3: Accuracy-cost frontiers at the 20% prefix. More data is not monotonically better in this small sample because each point retrains a finite diagnostic model; the conservative minimum-cost table therefore uses interval lower bounds rather than selecting by point estimate alone.

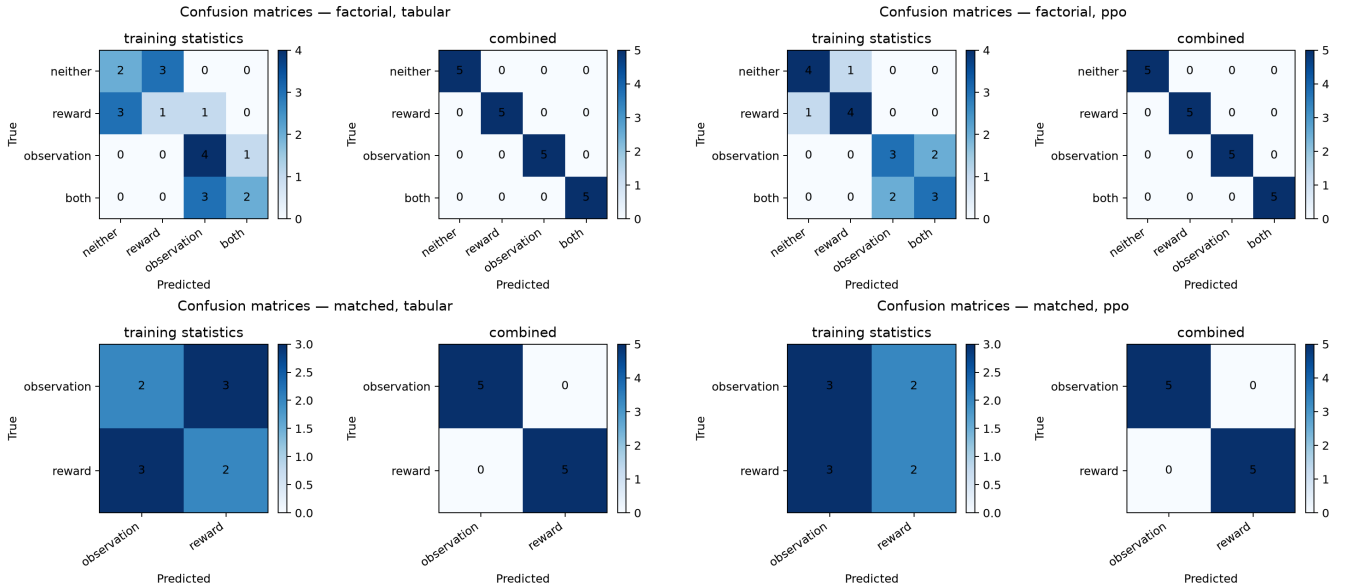


Figure 4: Leave-one-seed-out confusion matrices at the 20% prefix. Each panel contrasts standard passive statistics (left subpanel) with the 64-episode combined probe (right subpanel).

9 Shortcuts, confounds, and reproducibility

The factorial benchmark intentionally exposes observation identity and coverage, so it should not be interpreted as proving unrestricted passive identification. The matched control removes that shortcut. A state-only probe can solve this structured family because the proxy-state relation is itself diagnostic; unrestricted four-class guarantees still require A1–A5 and may require both probes. The history-dependence feature is predictive association, not a causal proof of intrinsic missing information. Finally, five seeds are appropriate for pipeline validation, not precise scientific claims.

The repository includes the environment and learner implementations, strict passive schema, raw passive and privileged logs, probe logs, run manifest, oracle gaps, all diagnostic metrics and confidence intervals, minimum-cost table, figures, animation, tests, and this source. The full experiment is reproduced by `python -m identirl run`; the playback is produced by `python -m identirl animate`. Eleven automated tests cover Gymnasium compatibility, factorial semantics, passive equivalence, value gaps, schema isolation, both learners, interventions, prefix extraction, and seed-level evaluation.

10 Limitations and falsifiable claims

The labels depend on a declared intended reward; no statistical procedure can recover an unspecified human objective. Oracle policies may be approximated in larger environments, confounding optimization error with the value gaps. Moreover, diagnosis is relative to Θ : unrestricted POMDP families admit observationally equivalent explanations. The project should therefore test three falsifiable claims: (i) passive accuracy remains near chance on matched equivalence pairs; (ii) the appropriate single probe separates each pure failure class; and (iii) combined probes achieve reliable four-class diagnosis at lower cost than indiscriminate privileged data collection.

11 Conclusion

Reward misspecification and observation insufficiency are distinct population properties, but they need not be distinguishable from ordinary early-training logs. The correct theoretical baseline is therefore conditional: identify the label only under explicit structural assumptions, and otherwise choose the least costly intervention that separates cross-label data distributions.

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